Forensic Data Recovery Report

Signature-based file carving and entropy analysis of a raw disk image. Generated 02 Oct 2026, 18:13 UTC.

Case number	PC-2026-001
Examiner	Shajilin Benadict S
Organisation	Karunya Institute of Technology and Sciences
Purpose	Recovery of deleted or hidden data from unallocated space
Tool	PhoenixCarve 1.0

1. Evidence item

File	test_evidence.dd
Size	130,386 bytes (127.3 KB)
SHA-256 (acquired)	`85b42c71f6cfe208d081b90b7551bb6ff08cb040738f26e707abc91ba615f884`
SHA-256 (after analysis)	`85b42c71f6cfe208d081b90b7551bb6ff08cb040738f26e707abc91ba615f884`
MD5	f1eb6bb1985c7a4f0ad9f37de877a61b
Integrity	**Preserved.** Hashes before and after analysis match; the image was opened read-only.

2. Method

The image was memory-mapped read-only and searched for known file signatures (magic numbers) from a database of file types. For each hit the end of the file was located from a footer marker, a size field stored in the header, or the next known header, bounded by a maximum size per type.

Every carved candidate was then checked structurally: JPEG marker walk, PNG chunk CRCs, ZIP member CRCs, PE section table, BMP and SQLite headers. Candidates whose structure failed were marked damaged; signature hits that proved to be random byte matches (for example a stray 'MZ') were rejected.

Shannon entropy was computed for every 512-byte block. High-entropy areas not explained by a recovered file were reported as possible encrypted or packed data. SHA-256 and MD5 were recorded for every recovered file.

3. Summary of findings

- **11** files were recovered (1 BMP, 1 EXE, 1 GIF87, 3 JPG, 1 PDF, 2 PNG, 2 ZIP).
- **9** passed structural validation; **2** are damaged and **0** could not be verified.
- **7** signature matches were rejected as false positives.
- **1** region(s) of high-entropy data totalling 32.0 KB match no file signature and may be encrypted.
- **1** zero-filled region(s) totalling 15.5 KB, consistent with deliberate wiping.

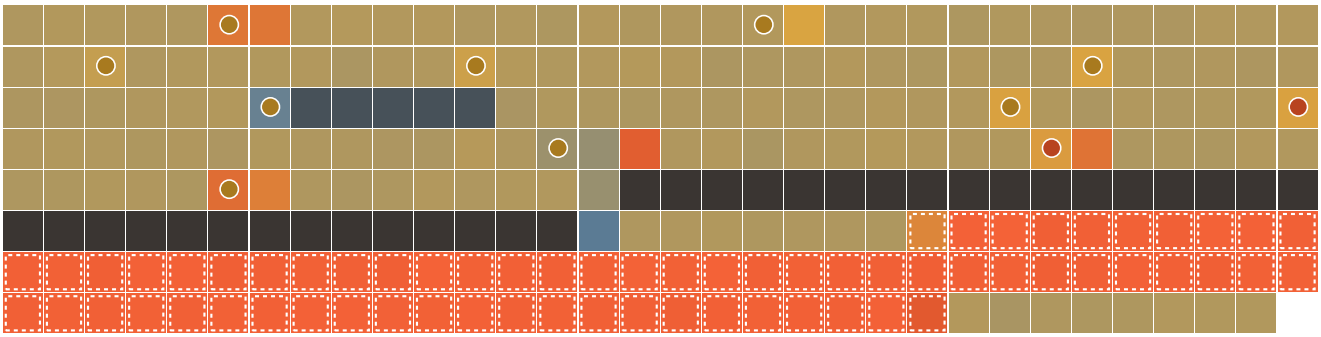

Figure 1. Entropy map, one cell per block, read left to right from offset 0. Dark is empty, blue is text, amber is binary, red is compressed or encrypted. Dots mark recovered files (amber valid, red damaged, blue unverified); dashed cells are unexplained high entropy.

4. Recovered files

#	File	Offset	Size	Status	Validation detail
1	`jpg_0000000a9e.jpg`	`0x00000a9e`	789 B	**valid**	EOI reached
2	`png_00000025fb.png`	`0x000025fb`	155 B	**valid**	all chunk CRCs OK
3	`pdf_0000004574.pdf`	`0x00004574`	125 B	**valid**	2 objects, trailer present
4	`zip_0000005604.zip`	`0x00005604`	154 B	**valid**	ZIP, 1 members, CRCs OK
5	`zip_0000007475.zip`	`0x00007475`	291 B	**valid**	DOCX, 2 members, CRCs OK
6	`bmp_0000008c6f.bmp`	`0x00008c6f`	3.1 KB	**valid**	DIB 40, size matches
7	`gif87_000000b03a.gif`	`0x0000b03a`	100 B	**valid**	40x40, trailer present
8	`png_000000bf49.png`	`0x0000bf49`	155 B	**corrupt**	CRC mismatch in IDAT chunk at +33
9	`exe_000000db43.exe`	`0x0000db43`	1.0 KB	**valid**	PE with 1 sections
10	`jpg_000000f34a.jpg`	`0x0000f34a`	6.5 KB	**corrupt**	expected marker at +609
11	`jpg_0000010a3d.jpg`	`0x00010a3d`	789 B	**valid**	EOI reached

5. Suspicious regions

Range	Size	Entropy	Assessment
`0x00016d00 to 0x0001ed00`	32.0 KB	7.59	No signature; possible encrypted container or packed data
`0x00011e00 to 0x00015c00`	15.5 KB	0.0	Zero-filled; consistent with wiping

7. Limitations

- Header/footer carving assumes files are stored contiguously. Fragmented files are recovered only up to the first break and are reported as damaged rather than repaired.
- Files with no structural validator are reported as unverified; their content was not confirmed.
- High entropy indicates compressed or encrypted data but does not prove encryption. Flagged regions require further examination.
- File names, timestamps and directory paths are not recoverable by carving; carved files are named by type and offset.

8. Chain of custody

Each entry stores the SHA-256 of the previous entry, so any later edit breaks the chain. Verification: **intact** (all entries intact).

#	Time (UTC)	Actor	Action	Entry hash
1	2026-10-02 18:13:09	Shajilin Benadict S	evidence received	`1dbc0d8248a836fb...`
2	2026-10-02 18:13:09	Shajilin Benadict S	carving completed	`20a4bab54ea60cb1...`
3	2026-10-02 18:13:09	Shajilin Benadict S	entropy analysis	`772158972ed79583...`
4	2026-10-02 18:13:09	Shajilin Benadict S	report generated	`ff0b1b92a342aa0c...`

9. Examiner declaration

I, Shajilin Benadict S, declare that the examination described in this report was carried out by me using PhoenixCarve 1.0 on a read-only copy of the evidence identified in section 1, that the evidence hash was unchanged by the examination, and that the findings are a true record of the results obtained.

Signature _____ Date _____

Appendix A. Hashes of recovered files

File	SHA-256
jpg_0000000a9e.jpg	82c96a6bb4f582ef8872c7d8cd6d6e0a35ed03057a294722ce20a4f19e7c7abf
png_00000025fb.png	a525c7de933154ef7612beaacf6de191e72854e011b7e38fd29e0c09ece26dff
pdf_0000004574.pdf	79370862c6cb54e96ed3125464c11e4ce3e8b08fb5048723695b6b9e9728d701
zip_0000005604.zip	3c7765b5f7e0b0902d99a9216a972ebf13406a9fa308903aedd86efdb655862b
zip_0000007475.zip	6c3109f5d635c63805d3fa6a0b4f5ed66901c4ce58735e6e77108728303bf6b5
bmp_0000008c6f.bmp	1958fa2d2f5ae945346aebd32412e6c6a4a792895585c52725497e8d5509d737
gif87_000000b03a.gif	4d5d99dce573f4b0067b6e777734dd33b322d57973ba5f78c5a4114e08cb1f6c
png_000000bf49.png	690ce986d48d226980916d9470f7def53f6960f6278bbd571a01d7ce42ce8a74
exe_000000db43.exe	cba12211949642fcc5e03820994d56dbd9e33f74e3fe36c23548082a076b05ba
jpg_000000f34a.jpg	67e6ac15dfe9198442c94e647307637e13635420ad7a05f98ce8aadcce41f575
jpg_0000010a3d.jpg	3355fd2565aaf6ea1008b361ec4a189a4be9d0df03b666b7be63b1173a833b2f